

UECM1034 ASSIGNMENT 1

Name:

Student ID:

COURSE CODE & COURSE TITLE: UECM1034 CALCULUS II

FACULTY: LKC FES, UTAR

COURSE: AM, AS, FM

SESSION: JUN 2026

LECTURER: LIEW HOW HUI

CLO4 Question	Mark(s)	CLO5 Question	Mark(s)
§A:I		§A:III	
§A:II		§A:IV	
§B:1		§B:2	
Penalty (if any)		Penalty (if any)	
CLO4 Total:	out of 10	CLO5 Total:	out of 10

Instructions:

1. **Write all your answers on A4 papers.** The A4 papers cannot be too thin and impossible for a scanner to scan.
2. **Your answers must be hand-written.** This is to prevent you from copying and pasting from the Internet. You need to make sure that you understand what you have written down because you may be randomly asked to present in the tutorial class. **If you fail to illustrate the answer you have written down on the whiteboard without looking as your answer,** you may receive zero mark for the question you are being asked.
3. Make sure your writings are not too tiny and **readable**. Terrible writings may receive penalty.
4. A **penalty** of up to 50% deduction of Section A if the submitted answers do not follow instructions 1. to 3.
5. **If you hand-writing for Section A and Section B differs significantly, penalty will be imposed on Section A.**
6. **Write your steps clearly.** Marks may be deducted or will not be awarded when non-obvious steps are skipped.
7. **Deadline for Section A submission is 2 pm on Wednesday of Week 6. Late submission will receive penalty and may not be eligible to take the Section B quiz.**
8. Section B is a **closed book one-hour quiz with two main questions (may have sub-questions)**. You are required to answer them just by using non-programmable calculator(s). Write down the answers for Section B on the space provided in the question paper and submit them to the lecturer within 1 hour.
9. If you forget to write down your name on the question paper, marks will be deducted.
10. **Don't Cheat!** Cheating in either Section A or Section B or both is a serious offence in violation of university regulations.

Section A (16%)

§A.I. (a) Find the values of $\frac{\partial^2 z}{\partial x^2}$ and $\frac{\partial^2 z}{\partial x \partial y}$ at $(x, y, z) = (6, 8, 43)$ given $xy + 5yz + xz - 2026 = 0$. (1 mark)

(b) Classify the critical points of the objective function an optimisation problem with an expression $f(x, y) = x^3 - 5x^2 + 5x - 5x(y + 1)^2$ (3 marks)

§A.II. (a) Let x, y, z be three sides of a triangle $\triangle$. Use optimisation method to find the maximum triangle area for the triangle $\triangle$ with the total length $x + y + z = 12$.

(b) Generalise the statement in part (a) with proper justification. (1 mark)

§A.III. (a) Find the exact value of the double integral

$$\int_0^2 \int_0^{\sqrt{4-x^2}} \sqrt{5-x^2-y^2} dy dx \quad (3 \text{ marks})$$

(b) State the Riemann sum for the integral in part (a) and explain how can you approximate the exact value in part (a) using Riemann sum. (1 mark)

§A.IV. Evaluate $\iiint_E \frac{z}{(4-y)^2} dV$ where E is the region lying in the first octant bounded by the cylinder $y^2 + z^2 = 4$ and the planes $z = 0$, $y = 0$, $x + y = 2$ and $x + 2y = 6$. (4 marks)

Section B (1 hour. 4%): The questions for Section B will be given during lecture in either Week 6 or Week 7. It is a closed book quiz with two main questions (with possibly multiple sub-questions). You are required to answer them just by using calculator(s).